HOW DOES MATHEMATICS CONTRIBUTE TO OUR UNDERSTANDING OF QUANTUM COMPUTATION?

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1 Introduction

The invention of the computer has undoubtedly been one of the greatest technological developments in human history. Over the years, the technology has grown exponentially as computers become more powerful all the while shrinking in size. According to Moore's Law, a computer's processing power doubles approximately every 18 months. However, this exponential growth is nearing its limits as the size of our processors become so small that electrons begin exhibiting quantum properties.

A computer is made up of transistors, electric switches controlling the flow of electrons. As transistors shrink to the size of mere atoms, electrons can transfer themselves across a blocked passage via quantum tunneling, which causes computation errors. Enter quantum physics, a scientific theory describing the behaviour of atoms and fundamental particles such as electrons and photons. Naturally then did the idea of quantum computing arise: to apply quantum properties such as superposition and entanglement to process information. It is this novel paradigm of computing that we will explore in this essay [1].

Quantum computing is grounded in mathematics, physics, and computer science. In this essay, we will examine the role mathematics plays in in this field.

Research Question: How does mathematics contribute to our understanding of quantum computation?

First, we will establish the prerequisite mathematical knowledge, namely linear algebra, before delving into the primitives that serve as the basis for quantum computational theory. Finally, an exploration of quantum algorithms will serve as a practical application of previous discussions in this essay.

2 Mathematical Background

In this first section, we investigate the mathematical knowledge, namely linear algebra, involved in quantum computational theory.

2.1 Dirac Notation, Vector Adjoint

While vector notation is commonly used in linear algebra, a different notation is seen by convention in quantum mechanics: *Dirac's Notation.*

In Dirac notation, column vectors are denoted by *kets*: $|\psi\rangle$. For example:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

For every ket, there exists a corresponding *bra* vector $\langle\psi|$, the *adjoint* of its ket.

$$\langle\psi| \equiv |\psi\rangle^\dagger$$

The adjoint of a vector $|v\rangle$, denoted with the symbol †, is a row vector whose entries are the complex conjugates of $|v\rangle$

$$\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}^\dagger = \begin{bmatrix} v_1^* & \dots & v_n^* \end{bmatrix}$$

The naming of a *bra* and a *ket* is derived from the inner product, denoted $\langle\psi|\psi\rangle$, which forms a *bra-ket* [2].

2.2 Hilbert Spaces

The mathematics of quantum computation is situated in *Hilbert Spaces*, which are complex vector spaces with an inner product [3].

2.2.1 Inner Products / Orthogonality

The inner product of 2 vectors $|u\rangle$ and $|v\rangle$, denoted $\langle u|v\rangle$, produces a scalar and is defined:

$$\begin{aligned} \langle u|v\rangle &= u^\dagger v \\ &= u_1^* v_1 + u_2^* v_2 + \cdots + u_n^* v_n \\ &= \sum_{k=0}^{n} (u_k^*) v_k \end{aligned}$$

The inner product satisfies the following properties. Given 2 vectors $|u\rangle$ and $|v\rangle$, and a scalar α then:

- $\langle u|v\rangle^* = \langle v|u\rangle$
- $\langle u|\alpha v\rangle = \alpha \langle u|v\rangle$
- $\langle v|v\rangle \geq 0$ with $\langle v|v\rangle = 0$ only if $|v\rangle = 0$

Two vectors $|u\rangle$ and $|v\rangle$ are *orthogonal* (perpendicular)if $\langle u|v\rangle = 0$ [4].

2.2.2 Norm / Orthonormality

The norm of a vector $|v\rangle$, denoted $||\,|v\rangle\,||$ is defined:

$$\begin{aligned} ||a|| &= \sqrt{\langle a|a\rangle} \\ &= \sqrt{\sum_{k=0}^{n-1}(a_k^*)a_k} \\ &= \sqrt{\sum_{k=0}^{n-1}|a_k|^2} \end{aligned}$$

A vector $|v\rangle$ is said to be *normalized* if $||\,|v\rangle\,|| = 1$. *Orthonormal* vectors are orthogonal unit vectors [5].

2.2.3 Linear Combinations / Basis

Vectors can be combined using scalar multiplication and vector addition to form new vectors.

For example, given two vectors $|u\rangle$ and $|v\rangle$, and two scalars a and b, a new vector $|w\rangle$ can be formed:

$$|w\rangle = a\,|u\rangle + b\,|v\rangle$$

Linear combinations can be thus generalized given a set of k vectors $\{|v_k\rangle\}$ and scalars $\{c_k\}$

$$\begin{aligned} |w\rangle &= c_0\,|v_0\rangle + c_1\,|v_1\rangle + c_2\,|v_2\rangle + \cdots + c_{n-1}\,|v_{n-1}\rangle \\ &= \sum_{k=0}^{n-1} c_0\,|v_k\rangle \end{aligned}$$

A *basis* of a vector space is a set of vectors, $\{|v_k\rangle\}$, such that any vector $|w\rangle$ in the vector space can be written as a linear combination of vectors in that set.

$$|w\rangle = \sum_i c_i\,|v_i\rangle$$

A basis has two characteristics:

- *Linear Independence* - Vectors are linearly independent if

  $$\sum_{i=1}^{n} \alpha_i\,|v_i\rangle = 0 \text{ if and only if } \alpha_i = 0$$

  That is, no basis vector in that set is a linear combination of other basis vectors.

- *Completeness* The set of vectors that form a basis must *span* the entire vector space [4].

The inner product of basis vectors $\{|v_1\rangle, \ldots, |v_n\rangle\}$ satisfy

$$\langle v_i|v_j\rangle = \delta_{ij}$$

Thus, for a vector

$$|w\rangle = \sum_i c_i\,|v_i\rangle$$

Its scalars can be determined with the inner product:

$$\begin{aligned}
|w\rangle &= \sum_i c_i |v_i\rangle \\
\langle v_k|w\rangle &= \left\langle v_k \middle| \sum_i c_i v_i \right\rangle \\
&= \sum_i c_i \langle v_k|v_i\rangle \\
&= \sum_i c_i \delta_{ki} \\
&= c_k
\end{aligned}$$

Therefore

$$c_k = \langle v_k|w\rangle$$

[6]

2.2.4 Completeness Relation

The completeness relation states that:

$$I = \sum_i |v_i\rangle \langle v_i|$$

Given $|v_i\rangle$ to be an orthonormal basis in V, any vector $|w\rangle$ is a linear combination of basis vectors $|v_i\rangle$:

$$|w\rangle = \sum_i c_i |v_i\rangle$$

Since $c_i = \langle v_i|w\rangle$, therefore:

$$\begin{aligned}
(\sum_i |v_i\rangle \langle v_i|) |w\rangle &= \sum_i |v_i\rangle \langle v_i|w\rangle \\
&= \sum_i c_i |v_i\rangle \\
&= |w\rangle
\end{aligned}$$

for any $|w\rangle$. Thus

$$\begin{aligned}
(\sum_i |v_i\rangle \langle v_i|) |w\rangle &= I |w\rangle \\
\sum_i |v_i\rangle \langle v_i| &= I
\end{aligned}$$

[3]

2.2.5 Outer Product

The outer product is the matrix multiplication of a ket and a bra, denoted $|u\rangle \langle v|$ [7].

The matrix $P_k = |v\rangle \langle v|$ is the projection onto the one-dimensional space defined by $|v\rangle$ [8].

2.2.6 Matrix Multiplication

Given an $n \times p$ matrix A and an $p \times m$ matrix B, then $C = AB$ is an $n \times m$ matrix whose ij^{th}element is given by the inner product of the corresponding row and column of matrix A and B respectively.

$$C_{ij} = (AB)_{ij} \equiv \sum_{k=1}^{n} A_{ik} B_{kj}$$

where $1 \leq i \leq n$ and $1 \leq j \leq p$.
This definition also applies when a matrix is multiplied by a vector (an $n \times 1$ matrix)

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{21} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_1 + \cdots + a1nx_1 \\ a_{21}x_2 + a_{22}x_2 + \cdots + a2nx_2 \\ \vdots \\ a_{m1}x_m + a_{m2}x_m + \cdots + a_{mn}x_n \end{bmatrix}$$

[7]

2.2.7 Linear Transformations

A linear transformation $T : V \to W$ is a function that maps a vector from vector space V to a different vector space W. Linear transformations hold the following properties:

$$T(c\,|v\rangle) = cT(|v\rangle)$$
$$T(|v_1\rangle + |v_2\rangle) = T(|v_1\rangle) + T(|v_2\rangle)$$

where c is a scalar.

Linear transformations are represented by matrices [4]. Given

$$|w\rangle = \sum_j c_j\,|v_j\rangle$$

$$\begin{aligned} T\,|w\rangle &= T \sum_j c_j\,|v_j\rangle \\ &= \sum_j c_j (T\,|v_j\rangle) \end{aligned}$$

We observe that the result of transformation T is determined by its action on the basis vectors.

Thus, for $T : V \to W$, and its respective vector basis $\{|v_1\rangle, \dots, |v_n\rangle\}$ and $\{|w_1\rangle, \dots, |w_m\rangle\}$,

$$T\,|v_j\rangle = \sum_{i=1}^{m} \sum_{j=1}^{n} \alpha_{ij}\,|w_i\rangle$$

the entries α_{ij} are the entries of $m \times n$ matrix T [3].

Using outer product notation:

$$\begin{aligned} T\,|v_j\rangle &= \sum_{i=1}^{m} \sum_{j=1}^{n} \alpha_{ij}\,|w_i\rangle \\ T &= \sum_{i=1}^{m} \sum_{j=1}^{n} \alpha_{ij}\,|w_i\rangle\,\langle v_j| \end{aligned}$$

[8] We prove that a transformation T on an arbitrary vector $|w\rangle$ is given by matrix multiplication:

$$\begin{aligned}
T|w\rangle &= \left(\sum_{i=1}^{m}\sum_{j=1}^{n}\alpha_{ij}|w_i\rangle\langle v_j|\right)\sum_{j=1}^{n}c_j|v_j\rangle \\
&= \sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=1}^{n}\alpha_{ij}c_k|w_i\rangle\langle v_j|v_j\rangle \\
&= \sum_{i=1}^{m}\sum_{j=1}^{n}\alpha_{ij}c_j|w_i\rangle(1) \\
&= \sum_{i=1}^{m}\sum_{j=1}^{n}\alpha_{ij}c_j|w_i\rangle \\
&= \sum_{j}c_j(T|v_j\rangle)
\end{aligned}$$

[9]

2.2.8 Matrix Transpose

The transpose of a matrix M is a new matrix M^t whose rows are the columns of M and columns the rows of M. In other words, if M is a $n \times m$ matrix, then M^t is a $m \times n$ matrix defined by:

$$(M_{ij})^t = (M_{ji})$$

[8]

2.2.9 Matrix Adjoint

The *adjoint* of a matrix M, denoted $M^\dagger$, is the conjugate transpose.

$$\begin{aligned}
M^\dagger &= (M^t)^* \\
M^\dagger_{ij} &= M^*_{ji}
\end{aligned}$$

[8]

2.2.10 Unitary Matrices

A matrix U is unitary if

$$UU^\dagger = U^\dagger U = 1$$

Unitary matrices preserve the inner product. That is,

$$\begin{aligned}
\langle Uv|Uv\rangle &= \langle v|U^\dagger U|v\rangle \\
&= \langle v|1|v\rangle \\
&= \langle v|v\rangle
\end{aligned}$$

[4]

2.2.11 Hermitian Matrices

A matrix U is Hermitian if

$$U^\dagger = U$$

[3]

2.2.12 Identity Matrix

A square matrix whose entries on the main diagonal are ones and zeroes elsewhere is known as the *identity matrix*.

$$I_{ij} = \delta_{ij}$$

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

The unit matrix is also the multiplicative identity: multiplying any matrix by I leaves the matrix unchanged [4].

3 Axioms of Quantum Mechanics

Underlying quantum computational theory is a set of axioms connecting the physical world and the mathematical framework of quantum mechanics [5].

3.1 Superposition Principle

Associated to any physical system is a Hilbert space, the state space of the system. The state of a closed system is described by its state vector, a unit/normalized vector in the state space [10].

3.1.1 The Qubit

The simplest, and fundamental quantum system is the 2-dimensional Hilbert space: the quantum bit, or *qubit*.

First, we define an orthonormal basis: the *computational/standard* basis states:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The basis states $|0\rangle$ and $|1\rangle$ are analogous to a classical bit's two possible states, 0 *or* 1, respectively. However, from the superposition principle, a qubit, being an arbitrary vector state in the space, can be a linear combination of basis states $|0\rangle$ and $|1\rangle$. That is, a qubit can be in both states 0 *and* 1 simultaneously!

An arbitrary qubit: $|\psi\rangle$ is denoted

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

where α and β are complex numbers denoting the respective *amplitudes* of the states $|0\rangle$ and $|1\rangle$. Since $|\psi\rangle$ is normalized:

$$\sqrt{\langle\psi|\psi\rangle} = 1$$

or

$$\begin{aligned} \sqrt{\langle\psi|\psi\rangle} &= 1 \\ \sqrt{(\alpha^*)\alpha + (\beta^*)\beta} &= 1 \\ (\alpha^*)\alpha + (\beta^*)\beta &= 1 \\ |\alpha|^2 + |\beta|^2 &= 1 \end{aligned}$$

$|\psi\rangle$ is said to be in a superposition of the states $|0\rangle$ and $|1\rangle$ with respective amplitudes α and β.

The state of a qubit $|\psi\rangle$ can be equally written using vector notation:

$$\begin{aligned} |\psi\rangle &= \alpha\,|0\rangle + \beta\,|1\rangle \\ &= \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \alpha \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \beta \end{bmatrix} \\ &= \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \end{aligned}$$

[11]

3.2 Unitary Evolution

The second postulate describes the evolution of the state $|\psi\rangle$ of a system with time.

The evolution of a closed quantum system is described by a unitary transformation. The state $|\psi'\rangle$ at time t_2 is related to the state $|\psi\rangle$ at time t_1 by some unitary transformation U [10].

$$|\psi'\rangle = U|\psi\rangle \longrightarrow UU^\dagger = I$$

3.2.1 Single Qubit Gates

A single qubit gate is a 2×2 unitary matrix manipulating a qubit's state.

X gate

The X gate, described by the following matrix

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

is the quantum analog of the classical NOT gate: $|0\rangle$ becomes $|1\rangle$ and $|1\rangle$ becomes $|0\rangle$.

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

The X gate on an arbitrary state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ produces the following:

$$X|\psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}$$

$$\alpha|0\rangle + \beta|1\rangle \xrightarrow{X} \alpha|1\rangle + \beta|0\rangle$$

We show that X is a unitary transformation: $XX^\dagger = 1$.

$$\begin{aligned} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= I \end{aligned}$$

Hadamard (H) gate

The Hadamard gate, H, is elementary in quantum computation.

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Applying the Hadamard gate on the standard basis states result in:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

Once again, we prove that $HH^\dagger = I$

$$\begin{aligned} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= I \end{aligned}$$

Z gate

The Z-gate changes a qubit's phase by negating the amplitude of $|1\rangle$.

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

For an arbitrary state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$:

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}$$

[12].

$$\begin{aligned} ZZ^\dagger &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= I \end{aligned}$$

3.3 Measurement Principle

The third postulate describes the consequences of observing a quantum system.

Quantum measurements are described by a collection of measurement operators $\{M_m\}$, where m refers to the measurement outcome. Given a quantum state $|\psi\rangle$, the probability that m occurs is:

$$p(m) = \langle\psi|\, M_m^\dagger M_m\, |\psi\rangle$$

and the state of the system after the measurement is:

$$\frac{M_m\,|\psi\rangle}{\sqrt{\langle\psi|\, M_m^\dagger M_m\, |\psi\rangle}}$$

Measurement operators are simply projections onto basis states.

$$M_m = |m\rangle\,\langle m|$$

Thus, for a qubit $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, its measurement operators in basis states $|0\rangle$ and $|1\rangle$ are:

$$M_0 = |0\rangle\,\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$M_1 = |1\rangle\,\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Measurement operators satisfy, $M_m^2 = M_m$. We prove the latter with the following result:

For a set of basis vectors $\{|v_1\rangle \ldots |v_k\rangle\}$ we show that the set of projectors $\{P_k = |v_k\rangle\,\langle v_k|\}$ satisfies $(P_k)^2 = P_k$:

$$\begin{aligned} (P_k)^2 &= (|v_k\rangle\,\langle v_k|)(|v_k\rangle\,\langle v_k|) \\ &= |v_k\rangle\,\langle v_k|v_k\rangle\,\langle v_k| \\ &= |v_k\rangle\,1\,\langle v_k| \\ &= |v_k\rangle\,\langle v_k| \\ &= P_k \end{aligned}$$

Furthermore, measurement operators are Hermitian: $M_m^\dagger = M_m$. We can thus simplify:

$$\begin{aligned} p(m) &= \langle\psi| M_m^\dagger M_m |\psi\rangle \\ &= \langle\psi| M_m M_m |\psi\rangle \\ &= \langle\psi| M_m |\psi\rangle \\ &= \langle\psi|m\rangle \langle m|\psi\rangle \end{aligned}$$

The probability of obtaining the outcome 0 for $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is:

$$\begin{aligned} p(0) &= \langle\psi| M_0^\dagger M_0 |\psi\rangle \\ &= \langle\psi|0\rangle \langle 0|\psi\rangle \\ &= \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \\ &= (\alpha^*)(\alpha) \\ &= |\alpha|^2 \end{aligned}$$

The state after measuring 0 becomes:

$$\begin{aligned} |\psi\rangle &= \frac{M_0 |\psi\rangle}{\sqrt{\langle\psi| M_0^\dagger M_0 |\psi\rangle}} \\ &= \frac{M_0 |\psi\rangle}{\sqrt{p(0)}} \\ &= \frac{1}{|\alpha|} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \\ &= \frac{1}{|\alpha|} \begin{bmatrix} \alpha \\ 0 \end{bmatrix} \\ &= \frac{\alpha}{|\alpha|} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \frac{\alpha}{|\alpha|} |0\rangle \\ &= |0\rangle \end{aligned}$$

Similarly, the probability of measuring 1 is $p(1) = |\beta|^2$, and the state after measuring 1 becomes $|1\rangle$.

Measurement operators also satisfy the completeness equation.

$$\sum_m M_m^\dagger M_m = I.$$

This means that probabilities sum to one:

$$\begin{aligned}
\sum_m p(m) &= \sum_m \langle\psi| M_m^\dagger M_m |\psi\rangle \\
&= \sum_m M_m^\dagger M_m \\
&= \sum_m M_m M_m \\
&= \sum_m (M_m)^2 \\
&= \sum_m M_m \\
&= M_0 + M_1 \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
&= I
\end{aligned}$$

[5]

Simply put, for a qubit

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

measurement probabilities of $|0\rangle$ and $|1\rangle$ are given by the squares of their amplitudes $|\alpha|^2$ and $|\beta|^2$, where $|\alpha|^2 + |\beta|^2 = 1$

3.4 Tensor Products

The state space of a composite physical system is the tensor product of the state spaces of the component physical systems [10].

Given n qubits, $\{|\psi_1\rangle \dots |\psi_n\rangle\}$ their joint state is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$, where $\otimes$ denotes the *tensor product*. Thus, for a multi-qubit system, their joint state is determined by taking the tensor products of the individual qubits.

The tensor product combines vector spaces to form larger vector spaces. The operation is defined as follows: given an $m \times n$ matrix M and an $p \times q$ matrix, the tensor product produces a matrix P of dimensions $mp \times nq$ defined as follows:

$$\begin{aligned}
P &= M \otimes N \\
&= \begin{bmatrix} M_{11} & \dots & M_{1n} \\ & \ddots & \\ M_{m1} & \dots & M_{mn} \end{bmatrix} \otimes \begin{bmatrix} N_{11} & \dots & N_{1q} \\ & \ddots & \\ N_{p1} & \dots & M_{pq} \end{bmatrix} \\
&= \begin{bmatrix} M_{11}\begin{bmatrix} N_{11} & \dots & N_{1q} \\ & \ddots & \\ N_{p1} & \dots & M_{pq} \end{bmatrix} & \dots & M_{1n}\begin{bmatrix} N_{11} & \dots & N_{1q} \\ & \ddots & \\ N_{p1} & \dots & M_{pq} \end{bmatrix} \\ & \ddots & \\ M_{m1}\begin{bmatrix} N_{11} & \dots & N_{1q} \\ & \ddots & \\ N_{p1} & \dots & M_{pq} \end{bmatrix} & \dots & M_{mn}\begin{bmatrix} N_{11} & \dots & N_{1q} \\ & \ddots & \\ N_{p1} & \dots & M_{pq} \end{bmatrix} \end{bmatrix}
\end{aligned}$$

The space of an n-qubit system is $(\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2n}$: an n-qubit system occupies a 2^n dimensional space.

The computational basis for an n-qubit system can be represented either in binary notation, $\{0,1\}^n$, or in decimal form $|0\rangle \dots |2^n - 1\rangle$, whereby

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \dots |2^n - 1\rangle = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Thus, the state of an n-qubit system can be generalized as:

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle = \sum_{x=0}^{2^n-1} \alpha_x |x\rangle$$

The two notations are equivalent.

Intuitively, quantum systems hold exponentially more information than classical systems: an n qubit system holds 2^n amplitudes, whereas an n bit classical system holds n numbers [5].

3.4.1 CNOT Gate

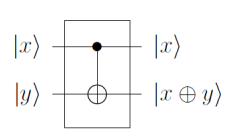
The CNOT (controlled-not) gate is an elementary operation in quantum computation acting on 2 qubits: a *control* qubit and a *target* qubit. The CNOT gate flips the value of the target qubit if the value of the control qubit is 1.

$$\begin{aligned} |00\rangle &\to |00\rangle \\ |01\rangle &\to |01\rangle \\ |10\rangle &\to |11\rangle \\ |11\rangle &\to |10\rangle \end{aligned}$$

The CNOT Gate is equivalent to a controlled X-gate.

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

For a control bit $|x\rangle$ and target bit $|y\rangle$, the CNOT gate is shown in a quantum circuit:



[12]

4 Bloch Sphere

Thus far, the qubit has been described using a 2-dimensional Hilbert space. The *Bloch Sphere* represents a qubit in a physical, 3-dimensional space.

We have previously denoted the state of a qubit as

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

In polar form:

$$\begin{aligned}\psi &= r_0 e^{i\phi_0}\,|0\rangle + r_1 e^{i\phi_1}\,|1\rangle \\ &= e^{i\phi_0}\Big(r_0\,|0\rangle + r_1 e^{i(\phi_1-\phi_0)}\,|1\rangle\Big)\end{aligned}$$

$e^{i\phi_0}$, the qubit's global phase, is irrelevant as it does not affect measurement outcomes. Thus, we can ignore $e^{i\phi_0}$.

Let

$$\phi_1 - \phi_0 = \phi$$

And since $r_0^2 + r_1^2 = 1$

$$\begin{aligned}|\psi\rangle &= r_0\,|0\rangle + r_1 e^{i(\phi_1-\phi_0)}\,|1\rangle \\ &= cos\theta\,|0\rangle + e^{i\phi} sin\theta\,|1\rangle\end{aligned}$$

Now, let

$$|\psi\rangle = cos\theta' + e^{i\phi} sin\theta'$$

For $\theta' = 0$, then $|\psi\rangle = 0$ and for $\theta' = \frac{\pi}{2}$, $|\psi\rangle = 1$. Thus, $0 \leq \theta' \leq \frac{\pi}{2}$ can represent every point on the Bloch sphere.

Furthermore, the opposite state of $|\psi\rangle$, which we denote $|\psi'\rangle$, will have polar coordinates $(1, \pi - \theta', \pi + \phi)$

$$\begin{aligned}|\psi'\rangle &= cos(\pi - \theta')\,|0\rangle + e^{i(\pi+\phi)} sin(\pi - \theta')\,|1\rangle \\ &= -cos(\theta')\,|0\rangle + e^{i\phi} e^{i\pi} sin(\theta')\,|1\rangle \\ &= -cos(\theta')\,|0\rangle - e^{i\phi} sin(\theta')\,|1\rangle \\ &= -\,|\psi\rangle\end{aligned}$$

Thus, opposite points differ only by a relative phase of -1 and are equivalent in the Bloch Sphere representation. In this manner, any point on the upper hemisphere of the Bloch sphere is defined:

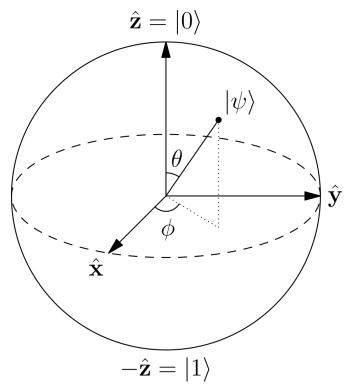
$$\theta = 2\theta' \longrightarrow \theta' = \frac{\theta}{2}$$

which results in the state

$$|\psi\rangle = cos\frac{\theta}{2}\,|0\rangle + e^{i\phi} sin(\frac{\theta}{2})\,|1\rangle$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$.

This is the *Bloch Sphere* representation of a qubit



The positive $|z\rangle$ axis corresponds to $\theta = 0$ and $\phi = 0$, which is effectively the $|0\rangle$ state. The state $-|z\rangle$ corresponds to $\theta = \pi$ and $\phi = 0$, which is the state $|1\rangle$. Interestingly, while $|0\rangle$ and $|1\rangle$ were orthogonal vectors in the 2-dimensional vector space, they are anti-polar states on the Bloch Sphere. The positive $|x\rangle$ axis corresponds to $\theta = \frac{\pi}{2}$ and $\phi = 0$, which is effectively the $|+\rangle$ state [13].

5 Quantum Phenomena

5.1 No Cloning Theorem

The No Cloning Theorem states that given an arbitrary unknown quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, it is impossible to make a copy of that quantum state. Mathematically, there exists no unitary operator U such that:

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$$

We assume that such a transformation U exists. If such is the case:

$$\begin{aligned} U(|0\rangle \otimes |0\rangle) &= |0\rangle \otimes |0\rangle \\ &= |00\rangle \end{aligned}$$

$$\begin{aligned} U(|1\rangle \otimes |0\rangle) &= |1\rangle \otimes |1\rangle \\ &= |11\rangle \end{aligned}$$

A contradiction soon arises:

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$$

$$\begin{aligned} LHS &= U(|\psi\rangle \otimes |0\rangle) \\ &= U\Big[(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle\Big] \\ &= U(\alpha|00\rangle + \beta|10\rangle) \\ &= \alpha U|00\rangle + \beta U|10\rangle \\ &= \alpha|00\rangle + \beta|11\rangle \end{aligned}$$

$$\begin{aligned} RHS &= |\psi\rangle \otimes |\psi\rangle \\ &= (\alpha|0\rangle + \beta|1\rangle) \otimes (\alpha|0\rangle + \beta|1\rangle) \\ &= \alpha^2|00\rangle + \alpha\beta|01\rangle + \alpha\beta|10\rangle + \beta^2|11\rangle \end{aligned}$$

$$\begin{aligned} \alpha|00\rangle + \beta|11\rangle &= \alpha^2|00\rangle + \alpha\beta|01\rangle + \alpha\beta|10\rangle + \beta^2|11\rangle \\ RHS &\neq LHS \end{aligned}$$

Such a contradiction proves that a quantum state cannot be cloned [14].

5.2 Entanglement

From the fourth postulate of quantum mechanics, the joint state of 2 individual qubits is given by their tensor products. However, *entangled* states are special states that cannot be factored. For example:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Assuming that $|\Phi^+\rangle$ can be written as the composite state of 2 qubits,

$$\begin{aligned} |\psi_0\rangle &= \alpha_0|0\rangle + \beta_0|1\rangle \\ |\psi_1\rangle &= \alpha_1|0\rangle + \beta_1|1\rangle \end{aligned}$$

Then,

$$\begin{aligned} |\psi_0\rangle \otimes |\psi_1\rangle &= (\alpha_0|0\rangle + \beta_0|1\rangle) \otimes (\alpha_1|0\rangle + \beta_1|1\rangle) \\ &= \alpha_0\beta_0|00\rangle + \alpha_0\beta_1|01\rangle + \alpha_1\beta_0|10\rangle + \alpha_1\beta_1|11\rangle \end{aligned}$$

The following equations are then necessarily true:

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \alpha_0\beta_0 |00\rangle + \alpha_0\beta_1 |01\rangle + \alpha_1\beta_0 |10\rangle + \alpha_1\beta_1 |11\rangle$$

$$\alpha_0\beta_0 = \frac{1}{\sqrt{2}} \tag{1}$$
$$\alpha_0\beta_1 = 0 \tag{2}$$
$$\alpha_1\beta_0 = 0 \tag{3}$$
$$\alpha_1\beta_1 = \frac{1}{\sqrt{2}} \tag{4}$$

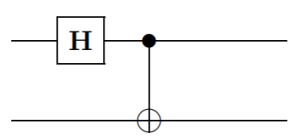
To satisfy (1) and (4), $\alpha_0, \beta_0, \alpha_1, \beta_1$ must be non-zero, in which case (2) and (3) no longer holds. The state $|\Phi^+\rangle$ cannot be factored.

When 2 qubits are entangled, the state of each individual qubit cannot be determined; rather it is their relationship that is of importance. Measuring $|\Phi^+\rangle$, the outcome is $|00\rangle$ with probability $\frac{1}{2}$ and $|11\rangle$ with equal probability. That is, if we measure the first qubit to be $|0\rangle$ or $|1\rangle$, then the second qubit collapses to $|0\rangle$ or $|1\rangle$ respectively with certainty, even if the two qubits are at opposite ends of the universe. This is precisely Einstein's notion of "spooky action at a distance."

The state $|\Phi^+\rangle$ is one of four *Bell states*, which exhibit maximum entanglement.

$$\begin{aligned}
\left|\Phi^+\right\rangle &= \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \\
\left|\Phi^-\right\rangle &= \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle \\
\left|\Psi^+\right\rangle &= \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle \\
\left|\Psi^-\right\rangle &= \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle
\end{aligned}$$

To generate the 4 Bell states, we apply a Hadamard gate to the first qubit followed by a CNOT targeting the second qubit.



For example:

$$\begin{aligned}
(H|0\rangle)|0\rangle &= (\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle)|0\rangle \\
&= \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle
\end{aligned}$$

Then, applying a CNOT:

$$\begin{aligned}
\frac{1}{\sqrt{2}}|0\rangle|0 \oplus 0\rangle + \frac{1}{\sqrt{2}}|1\rangle|1 \oplus 0\rangle &= \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \\
&= \left|\Phi^+\right\rangle
\end{aligned}$$

5.3 Superdense Coding

Superdense Coding is a quantum protocol that sends 2 bits of information with only one bit. We denote Alice to be the sender, and Bob the receiver.

We assume that Alice and Bob share a Bell state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0_A 0_B\rangle + |1_A 1_B\rangle)$$

where $_A$ and $_B$ denote Alice's and Bob's respective share of the Bell state.

Depending on Alice's two bits of information, she applies the following gates to her share of the Bell state.

- $00 \rightarrow$ identity operation
- $01 \rightarrow X$ gate
- $10 \rightarrow Z$ gate
- $11 \rightarrow XZ$ gate

Alice then sends her qubit to Bob, who un-entangles the qubits by applying a CNOT gate with Alice's qubit as the control bit and his qubit the target bit. Bob then applies the Hadamard transform and measures both qubits to recover Alice's 2-bit message.

Bits	*Alice's Gates*		*Bob's CNOT*	*Bob's Hadamard*	*Bob's Measurement*
00	I	$\frac{1}{\sqrt{2}}(\|0_A 0_B\rangle + \|1_A 1_B\rangle)$	$\frac{1}{\sqrt{2}}(\|0_A\rangle + \|1_A\rangle)\|0_B\rangle$	$\|0_A\rangle\|0_B\rangle$	$\|00\rangle$
01	X	$\frac{1}{\sqrt{2}}(\|1_A 0_B\rangle + \|0_A 1_B\rangle)$	$\frac{1}{\sqrt{2}}(\|1_A\rangle + \|0_A\rangle)\|1_B\rangle$	$\|0_A\rangle\|1_B\rangle$	$\|01\rangle$
10	Z	$\frac{1}{\sqrt{2}}(\|0_A 0_B\rangle - \|1_A 1_B\rangle)$	$\frac{1}{\sqrt{2}}(\|0_A\rangle - \|1_A\rangle)\|0_B\rangle$	$\|1_A\rangle\|0_B\rangle$	$\|10\rangle$
11	XZ	$\frac{1}{\sqrt{2}}(\|1_A 0_B\rangle - \|0_A 1_B\rangle)$	$\frac{1}{\sqrt{2}}(\|1_A\rangle - \|0_A\rangle)\|1_B\rangle$	$-\|1_A\rangle\|1_B\rangle$	$\|11\rangle$

[4]

5.4 Quantum Teleportation

Quantum teleportation is a protocol allowing Alice to send a quantum state $|\psi\rangle$ to Bob.

Alice and Bob must share a Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0_A 0_B\rangle + |1_A 1_B\rangle)$$

We denote $|\phi_n\rangle$ as the state of the whole quantum system, during the nth step of the protocol.

We begin with the qubit,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

that Alice wishes to teleport, and a Bell state, $|\Phi^+\rangle$.

$$\begin{aligned}
|\phi_0\rangle &= |\psi\rangle|\Phi^+\rangle \\
&= (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0_A 0_B\rangle + |1_A 1_B\rangle) \\
&= \frac{1}{\sqrt{2}}\Big[\alpha|0\rangle(|0_A 0_B\rangle + |1_A 1_B\rangle) + \beta|1\rangle(|0_A 0\rangle + |1_B 1\rangle)\Big]
\end{aligned}$$

Alice applies a CNOT gate with her qubit as the control bit and her share of the Bell state as the target qubit.

$$
\begin{aligned}
|\phi_1\rangle &= CNOT(|\phi_0\rangle) \\
&= CNOT\frac{1}{\sqrt{2}}\Big[\alpha\,|0\rangle\,(|0_A0_B\rangle + |1_A1_B\rangle) + \beta\,|1\rangle\,(|0_A0\rangle + |1_B1\rangle)\Big] \\
&= \frac{1}{\sqrt{2}}\Big[\alpha\,|0\rangle\,\Big(\,|(0\oplus 0_A)0_B\rangle + |(0\oplus 1_A)1_B\rangle\,\Big) \\
&\qquad + \beta\,|1\rangle\,\Big(\,|(0_A\oplus 1)0_B\rangle + |(1_A\oplus 1)1_B\rangle\,\Big)\Big] \\
&= \frac{1}{\sqrt{2}}\Big[\alpha\,|0\rangle\,(|0_A0_B\rangle + |1_A1_B\rangle) + \beta\,|1\rangle\,(|1_A0_B\rangle + |0_A1_B\rangle)\Big]
\end{aligned}
$$

Alice then sends her qubit through a Hadamard gate.

$$
\begin{aligned}
|\phi_2\rangle &= \frac{1}{\sqrt{2}}\Big[H\alpha\,|0\rangle\,(|0_A0_B\rangle + |1_A1_B\rangle) + H\beta\,|1\rangle\,(|1_A0_B\rangle + |0_A1_B\rangle)\Big] \\
&= \frac{1}{\sqrt{2}}\Big[\frac{\alpha}{\sqrt{2}}(|0\rangle + |1\rangle)(|0_A0_B\rangle + |1_A1_B\rangle) \\
&\qquad + \frac{\beta}{\sqrt{2}}(|0\rangle - |1\rangle)(|1_A0_B\rangle + |0_A1_B\rangle)\Big] \\
&= \frac{1}{2}\Big[\alpha(|0\rangle + |1\rangle)(|0_A0_B\rangle + |1_A1_B\rangle) \\
&\qquad + \beta(|0\rangle - |1\rangle)(|1_A0_B\rangle + |0_A1_B\rangle)\Big]
\end{aligned}
$$

The state expression can be rearranged to separate Alice's and Bob's qubits through factoring.

$$
\begin{aligned}
|\phi_2\rangle &= \frac{1}{2}\Big[\alpha\,|00_A0_B\rangle + \alpha\,|10_A0_B\rangle + \alpha\,|01_A1_B\rangle + \alpha\,|11_A1_B\rangle \\
&\qquad + \beta\,|01_A0_B\rangle - \beta\,|11_A0_B\rangle + \beta\,|00_A1_B\rangle - \beta\,|10_A1_B\rangle\Big] \\
&= \frac{1}{2}\Big[\,|00_A\rangle\,(\alpha\,|0_B\rangle + \beta\,|1_B\rangle) + |01_A\rangle\,(\alpha\,|1_B\rangle + \beta\,|0_B\rangle) \\
&\qquad + |10_A\rangle\,(\alpha\,|0_B\rangle - \beta\,|1_B\rangle) + |11_B\rangle\,(\alpha\,|1_B\rangle - \beta\,|0_B\rangle)\Big]
\end{aligned}
$$

If Alice performs a measurement, Bob's qubit results in one of four possible states.

Alice's Measurement Outcome	*Bob's Qubit's Resulting State*
$\|00\rangle$	$\alpha\,\|0\rangle + \beta\,\|1\rangle$
$\|01\rangle$	$\alpha\,\|1\rangle + \beta\,\|0\rangle$
$\|10\rangle$	$\alpha\,\|0\rangle - \beta\,\|1\rangle$
$\|11\rangle$	$\alpha\,\|1\rangle - \beta\,\|0\rangle$

Bob can easily recover the qubit $|\psi\rangle$, depending on Alice's outcome:

- Alice's measurement: $|00\rangle \longrightarrow$ Bob leaves his qubit alone.
- Alice's measurement: $|01\rangle \longrightarrow$ Bob applies the X gate.

$$X(\alpha\,|1\rangle + \beta\,|0\rangle) = \alpha\,|0\rangle + \beta\,|1\rangle$$

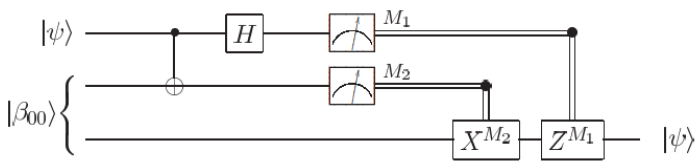
- Alice's measurement: $|10\rangle \longrightarrow$, Bob applies the Z gate.

$$X(\alpha\,|0\rangle - \beta\,|1\rangle) = \alpha\,|0\rangle + \beta\,|1\rangle$$

- Alice's measurement: $|11\rangle \longrightarrow$, Bob applies X and Z, Z gate.

$$X(\alpha\,|1\rangle - \beta\,|0\rangle) = \alpha\,|1\rangle - \beta\,|0\rangle$$
$$Z(\alpha\,|1\rangle - \beta\,|0\rangle) = \alpha\,|0\rangle + \beta\,|1\rangle$$

The quantum circuit for teleportation is shown:



The bottom qubit is Bob's share of the EPR [5].

6 Quantum Computation

6.1 Reversible Computation

From the second postulate of quantum mechanics, a transformation U on a quantum system must be unitary: $UU^\dagger = U^\dagger U = I$. Thus, there must exist an inverse operator $U^\dagger$ which reverses, or "undoes" a transformation U This is the notion of *reversible computation*. [15]

The latter is relevant in quantum algorithms, with the concept of an *oracle*. This is a a black box operation used to compute a function's output given its input. It is termed a black box because we cannot look inside the box to see the function's properties; rather we can only provide it with inputs and receive its outputs.

Instead of conceiving O as simply computing the function

$$O\,|x\rangle = |f(x)\rangle$$

it is defined as

$$O(|x\rangle \otimes |y\rangle) = |x\rangle\,|y \oplus f(x)\rangle$$

so that O is reversible [4].

6.2 Quantum Parallelism

Quantum parallelism is the idea that a quantum computer can simultaneous evaluate a function $f(x)$ for different values of x, and is the fundamental building block of numerous quantum algorithms.

Given a function $f(x) : \{0,1\} \to \{0,1\}$, we can evaluate this function using 2 qubits, initially in the state $|x, y\rangle$. With an oracle O, we can obtain $|x, y \oplus f(x)\rangle$. The first register, x, is termed the query register, and the second register, y is the answer register.

If we input

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

into our query register (by applying a Hadamard gate on $|0\rangle$), then applying U_f results in the following state:

$$\frac{|0, f(0)\rangle + |1, f(1)\rangle}{\sqrt{2}}$$

With only one single circuit, we obtain a state that contains information about both $f(0)$ and $f(1)$. This procedure can be applied to multi-qubit systems by applying a higher-order Hadamard transform. This operation applies n Hadamard gates on n qubits. The Hadamard transform on n qubits is denoted: $H^{\otimes n}$ (H tensored with itself n times).

In this sense, quantum parallelism enables us to compute all possible values of f to be evaluated simultaneously. However, this parallelism is not immediately useful, as to extract information from this state, we must perform a measurement which yields only $f(x)$ for a single value of x [5].

6.3 Hadamard Transform

As discussed in the previous section, the Hadamard Transform is at the heart of quantum parallelism.

Recall that the Hadamard Transform on a single qubit is a transformation that maps $|0\rangle \to |+\rangle$ and $|1\rangle \to |-\rangle$.

The Hadamard transform can also be performed on 2 qubits, in which case:

$$\begin{aligned}
H^{\otimes 2}\left|00\right\rangle &= \left|++\right\rangle \\
&= (\frac{1}{\sqrt{2}}\left|0\right\rangle + \frac{1}{\sqrt{2}}\left|1\right\rangle) \otimes (\frac{1}{\sqrt{2}}\left|0\right\rangle + \frac{1}{\sqrt{2}}\left|1\right\rangle) \\
&= \frac{1}{2}\left|00\right\rangle + \frac{1}{2}\left|01\right\rangle + \frac{1}{2}\left|10\right\rangle + \frac{1}{2}\left|11\right\rangle \\
H^{\otimes 2}\left|01\right\rangle &= \left|+-\right\rangle \\
&= (\frac{1}{\sqrt{2}}\left|0\right\rangle + \frac{1}{\sqrt{2}}\left|1\right\rangle) \otimes (\frac{1}{\sqrt{2}}\left|0\right\rangle - \frac{1}{\sqrt{2}}\left|1\right\rangle) \\
&= \frac{1}{2}\left|00\right\rangle - \frac{1}{2}\left|01\right\rangle + \frac{1}{2}\left|10\right\rangle - \frac{1}{2}\left|11\right\rangle \\
H^{\otimes 2}\left|10\right\rangle &= \left|-+\right\rangle \\
&= (\frac{1}{\sqrt{2}}\left|0\right\rangle - \frac{1}{\sqrt{2}}\left|1\right\rangle) \otimes (\frac{1}{\sqrt{2}}\left|0\right\rangle + \frac{1}{\sqrt{2}}\left|1\right\rangle) \\
&= \frac{1}{2}\left|00\right\rangle + \frac{1}{2}\left|01\right\rangle - \frac{1}{2}\left|10\right\rangle - \frac{1}{2}\left|11\right\rangle \\
H^{\otimes 2}\left|11\right\rangle &= \left|--\right\rangle \\
&= (\frac{1}{\sqrt{2}}\left|0\right\rangle - \frac{1}{\sqrt{2}}\left|1\right\rangle) \otimes (\frac{1}{\sqrt{2}}\left|0\right\rangle - \frac{1}{\sqrt{2}}\left|1\right\rangle) \\
&= \frac{1}{2}\left|00\right\rangle - \frac{1}{2}\left|01\right\rangle - \frac{1}{2}\left|10\right\rangle + \frac{1}{2}\left|11\right\rangle
\end{aligned}$$

Its matrix is:

$$\begin{aligned}
H^{\otimes 2} &= H \otimes H \\
&= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \otimes \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} & \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \\ \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} & \frac{-1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \end{bmatrix} \\
&= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}
\end{aligned}$$

The Hadamard transform on two qubits also outputs an equal superposition of basis states, all amplitudes being $\pm\frac{1}{2}$. The amplitude is negative when the input and output for a qubit are both $\left|1\right\rangle$. For example, for input $\left|11\right\rangle$, the resulting amplitudes of $\left|01\right\rangle$ and $\left|10\right\rangle$ are negative, but positive for $\left|11\right\rangle$.

We observe that the sign is determined by the parity of such cases where $\left|1\right\rangle \to \left|1\right\rangle$. In this manner, the Hadamard transform for n-qubits can be generalized. The Hadamard Transform on n $\left|0\right\rangle$ qubits:

$$\begin{aligned}
H^{\otimes n}\left|0\right\rangle^{\otimes n} &= (\frac{1}{\sqrt{2}}\left|0\right\rangle + \frac{1}{\sqrt{2}}\left|1\right\rangle)^{\otimes n} \\
&= \frac{1}{2^{n/2}} \sum_{x \in \{0,1\}^n} \left|x\right\rangle
\end{aligned}$$

The Hadamard transform on an arbitrary state $|u\rangle = |u_1 u_2 \dots u_n\rangle$, where $u_1 \dots u_n$ are the n-bits of $|u\rangle$:

$$H^{\otimes n} |u\rangle = \sum_x \frac{(-1)^{u \cdot x}}{2^{n/2}} |x\rangle$$

where $u \cdot x = u_1 x_1 + u_2 x_2 + \cdots + u_n x_n$. This dot product is effectively counting the parity of $|1\rangle \to |1\rangle$ transformations [11].

7 Quantum Algorithms

In this section, we will apply the knowledge acquired thus far in this essay to examine a series of algorithms.

The quantum algorithms described below assume the provision of an *oracle*, O, to compute a function $f(x)$.

$$O(|x\rangle |y\rangle) = O(|x\rangle |y \oplus f(x)\rangle)$$

We will also avoid any in-depth discussions about the speedup quantum algorithms provide over their classical counterparts, as such analysis permeates into the field of computer science.

We will denote $|\phi_n\rangle$ as the state of the quantum system for the nth step of the algorithm.

7.1 Deutsch-Jozsa Algorithm

The Deutsch-Jozsa Algorithm is one of the first examples of a quantum algorithm proven to be exponentially faster than a classical algorithm.

Problem statement:
Suppose we are given a boolean function $f : \{0,1\}^n \to \{0,1\}$. We are promised that the function is one of 2 possibilities:

1. f is constant: $f(x) = 0$ or $f(x) = 1$ for all $x \in \{0,1\}^n$.
2. f is balanced: the number of inputs $x \in \{0,1\}^n$ for which f outputs 0 and 1 are equal. That is, $f(x) = 0$ for 2^{n-1} inputs and $f(x) = 1$ for 2^{n-1} inputs.

The task is to determine which is the case.

Algorithm:
Classically, if f is balanced, it takes at worst $2^{n-1}+1$ queries. For instance, the oracle can output 2^{n-1} zeros before outputting a one. The Deutsch-Jozsa Algorithm solves this problem with only 1 query.

We initialize the quantum system in the following state:

$$|\phi_0\rangle = |0\rangle^{\otimes n} |1\rangle$$

where $|0\rangle^{\otimes n}$ is the query register and $|1\rangle$ the answer register. We perform the Hadamard transform on both registers to obtain the following:

$$\begin{aligned} |\phi_1\rangle &= H\Big(|0\rangle^{\otimes n} |1\rangle \Big) \\ &= \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle \left(\frac{|0\rangle - |1\rangle}{2} \right) \end{aligned}$$

Next, we query the function with a black box.

$$\begin{aligned} |\phi_2\rangle &= O\Big(|x\rangle^n \frac{|0\rangle - |1\rangle}{\sqrt{2}} \Big) \\ &= |x\rangle^n \frac{|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}} \end{aligned}$$

We can simplify this by considering the 2 cases for $f(x)$.

- When $f(x) = 0$:

$$
\begin{aligned}
|\phi_2\rangle &= |x\rangle^n \frac{|0 \oplus 0\rangle - |1 \oplus 0\rangle}{\sqrt{2}} \\
&= |x\rangle^n \frac{|0\rangle - |1\rangle}{\sqrt{2}}
\end{aligned}
$$

- When $f(x) = 1$:

$$
\begin{aligned}
|\phi_2\rangle &= |x\rangle^n \frac{|0 \oplus 1\rangle - |1 \oplus 1\rangle}{\sqrt{2}} \\
&= |x\rangle^n \frac{|1\rangle - |0\rangle}{\sqrt{2}} \\
&= -|x\rangle^n \frac{|0\rangle - |1\rangle}{\sqrt{2}}
\end{aligned}
$$

The state $|\phi_2\rangle$ differs by a phase of -1 for $f(x) = 1$. Thus, we can "write" the output of $f(x)$ into the phase of $|\phi_2\rangle$:

$$
|\phi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}
$$

Storing the value of $f(x)$ in a state's phase is known as *"phase kickback,"*, which consists of initializing the answer register as $|-\rangle$ before quering the oracle.

Next, we apply the Hadamard transform to the query register. The answer register is omitted hereon as it is no longer needed.

$$
\begin{aligned}
|\phi_3\rangle &= H^{\otimes n}\Big(\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle\Big) \\
&= \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \Big(\frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle\Big) \\
&= \sum_{y \in \{0,1\}^n} \Big(\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x) + x \cdot y}\Big) |y\rangle
\end{aligned}
$$

Finally, we examine the probability of measuring $|y\rangle = |0\rangle^{\otimes n}$. Since $|y\rangle$ is $|00\ldots0\rangle$, then $x \cdot y = 0$. Therefore:

$$
\begin{aligned}
p(|0\rangle^{\otimes n}) &= \Big|\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x) + x \cdot y}\Big|^2 \\
&= \Big|\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)}\Big|^2
\end{aligned}
$$

If $f(x)$ is balanced:

$$
\begin{aligned}
\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} &= \frac{1}{2^n} \sum_{1}^{2^{n-1}} \Big((-1)^0 + (-1)^1\Big) \\
&= 0
\end{aligned}
$$

Thus, if $f(x)$ is balanced, we should never observe the state $|0\rangle^{\otimes n}$.

On the other hand, if $f(x)$ is a constant function:

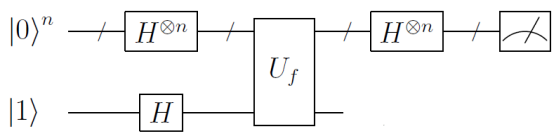
$$\begin{aligned} p(|0\rangle^{\otimes n}) &= \Big|\frac{1}{2^n}\sum_{x=0}^{2^n-1}(-1)^{f(x)}\Big|^2 \\ &= \begin{cases} \Big|\frac{1}{2^n}(2^n)\Big|^2 \text{ if } f(x)=0 \\ \Big|\frac{1}{2^n}(-2^n)\Big|^2 \text{ if } f(x)=1 \end{cases} \\ &= 1 \end{aligned}$$

The probability of measuring $|0\rangle^n$ is:

$$\Big|\frac{1}{2^n}\sum_{x\in\{0,1\}^n}(-1)^{f(x)}\Big|^2 = \begin{cases} 1 \text{ if } f(x) \text{ is constant} \\ 0 \text{ if } f(x) \text{ is balanced} \end{cases}$$

That is, if $f(x)$ is constant, we should always measure $|0\rangle^{\otimes n}$ and never $|0\rangle^{\otimes n}$ if $f(x)$ is balanced. With only one query, the Deutsch-Jozsa algorithm is able to determine whether f is constant or balanced [16].

The algorithm's circuit is as follows [4]:



7.2 Bernstein-Vazirani Algorithm

Problem statement:
Given a black box function $f : \{0,1\}^n \to \{0,1\}$, $f(x) = u \cdot x$ for some $u \in \{0,1\}^n$, find u.

Algorithm:
Classically, we are able to solve for one bit of u with each query: n queries are necessary [4].

$$\begin{aligned} f(100\ldots00) &= u_1 \\ f(010\ldots00) &= u_2 \\ f(001\ldots00) &= u_3 \\ &\vdots \\ f(000\ldots10) &= u_{n-1} \\ f(000\ldots01) &= u_n \end{aligned}$$

The Bernstein-Vazirani Algorithm solves for u with one query.

We initialize a system with 2 quantum registers, $|0\rangle^n$ and $|1\rangle$ to achieve *phase-kickback*:

$$|\phi_0\rangle = |0\rangle^n |1\rangle$$

Applying the Hadamard transform

$$\begin{aligned} |\phi_1\rangle &= H\Big(|0\rangle^{\otimes n}|1\rangle\Big) \\ &= \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle\Big(\frac{|0\rangle-|1\rangle}{2}\Big) \end{aligned}$$

Applying the Oracle function, O:

$$\begin{aligned}
|\phi_2\rangle &= O\Big[\frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle \Big(\frac{|0\rangle - |1\rangle}{2}\Big)\Big] \\
&= \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} |x\rangle \\
&= \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{u\cdot x} |x\rangle
\end{aligned}$$

[17] We decompose $|\phi_2\rangle$ as the tensor product of individual qubits:

$$\begin{aligned}
|\phi_2\rangle &= \frac{(-1)^{u_1\cdot 0}|0\rangle + (-1)^{u_1\cdot 1}|1\rangle}{\sqrt{2}} \otimes \cdots \otimes \frac{(-1)^{u_n\cdot 0}|0\rangle + (-1)^{u_n\cdot 1}|1\rangle}{\sqrt{2}} \\
&= \frac{|0\rangle + (-1)^{u_1}|1\rangle}{\sqrt{2}} \otimes \cdots \otimes \frac{|0\rangle + (-1)^{u_n}|1\rangle}{\sqrt{2}}
\end{aligned}$$

The qubits where $u_i = 1$ are in the state

$$\begin{aligned}
\frac{|0\rangle + (-1)^1|1\rangle}{\sqrt{2}} &= \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\
&= |-\rangle
\end{aligned}$$

The qubits where $u_i = 0$ are in the state

$$\begin{aligned}
\frac{|0\rangle + (-1)^0|1\rangle}{\sqrt{2}} &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\
&= |+\rangle
\end{aligned}$$

If we re-Hadamard the query register, for qubits where:

- $u_i = 1$
$$H|-\rangle = |1\rangle$$
- $u_i = 0$
$$H|+\rangle = |0\rangle$$

The qubits are exactly the state u. Thus, we measure the qubits to retrieve with certainty the n bits of u [18].

7.3 Simon's Algorithm

Problem statement:
Given a 2-to-1 function $f : \{0,1\}^n \to \{0,1\}^n$, such that $f(x) = f(x \oplus a)$, find a.

Algorithm:
The algorithm uses 2 registers, both initialized to the state $|0^n\rangle$.

$$|\phi_0\rangle = |0^n\rangle |0^n\rangle$$

We perform the Hadamard transform on the first register:

$$\begin{aligned}
|\phi_1\rangle &= (H^{\otimes n}|0\rangle^{\otimes n})|0\rangle^{\otimes n} \\
&= \frac{1}{2^{n/2}} \sum_{x\in\{0,1\}^n} |x\rangle |0\rangle^{\otimes n}
\end{aligned}$$

Next we apply the oracle function:

$$|\phi_2\rangle = \frac{1}{2^{n/2}} \sum_{x\in\{0,1\}^n} |x\rangle\,|f(x)\rangle$$

Upon measuring the second register, the second register collapses to an arbitrary, specific value of the function $f(x)$: $f(z)$. There are 2 values of x such that $f(x) = f(z)$: z and $z \oplus a$. Thus, the result of the measurement is:

$$|\psi_3\rangle = \Big(\frac{1}{\sqrt{2}}\,|z\rangle + \frac{1}{\sqrt{2}}\,|z\oplus a\rangle\Big)\,|f(z)\rangle$$

We perform a Hadamard transform on the first register. The second register $|f(z)\rangle$ is no longer useful and is omitted hereon.

$$\begin{aligned}
|\psi_4\rangle &= H^{\otimes n}\Big(\frac{1}{\sqrt{2}}\,|z\rangle + \frac{1}{\sqrt{2}}\,|z\oplus a\rangle\Big)\\
&= \frac{1}{\sqrt{2}}H^{\otimes n}\,|z\rangle + \frac{1}{\sqrt{2}}H^{\otimes n}\,|z\oplus a\rangle\\
&= \frac{1}{\sqrt{2}}\Big(\frac{1}{\sqrt{2^n}}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}\,|y\rangle\Big) + \frac{1}{\sqrt{2}}\Big(\frac{1}{\sqrt{2^n}}\sum_{y\in\{0,1\}^n}(-1)^{(z\oplus a)\cdot y}\,|y\rangle\Big)\\
&= \frac{1}{\sqrt{2^{n+1}}}\sum_{y\in\{0,1\}^n}\Big((-1)^{z\cdot y} + (-1)^{(z\oplus a)\cdot y}\Big)\,|y\rangle\\
&= \frac{1}{\sqrt{2^{n+1}}}\sum_{y\in\{0,1\}^n}\Big((-1)^{z\cdot y} + (-1)^{(z\cdot y)\oplus(a\cdot y)}\Big)\,|y\rangle\\
&= \frac{1}{\sqrt{2^{n+1}}}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}\Big(1 + (-1)^{a\cdot y}\Big)\,|y\rangle
\end{aligned}$$

We then calculate the amplitudes of the first register:

- When $a \cdot y = 1$:

$$\begin{aligned}
|\psi_4\rangle &= \frac{1}{\sqrt{2^{n+1}}}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}\Big(1 + (-1)^{a\cdot y}\Big)\,|y\rangle\\
&= \frac{1}{\sqrt{2^{n+1}}}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}\Big(1 + (-1)^{1}\Big)\,|y\rangle\\
&= \frac{1}{\sqrt{2^{n+1}}}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}0\,|y\rangle\\
&= 0\,|y\rangle
\end{aligned}$$

- When $a \cdot y = 0$:

$$\begin{aligned}
|\psi_4\rangle &= \frac{1}{\sqrt{2^{n+1}}}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}\Big(1 + (-1)^{a\cdot y}\Big)\,|y\rangle\\
&= \frac{1}{\sqrt{2^{n+1}}}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}\Big(1 + (-1)^{0}\Big)\,|y\rangle\\
&= \frac{2}{\sqrt{2^{n+1}}}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}\,|y\rangle\\
&= \frac{1}{\sqrt{2^{n-1}}}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}\,|y\rangle
\end{aligned}$$

Thus, the probabilities of measurements are:

$$\Big|\frac{1}{2^n}\sum_{y\in\{0,1\}^n}(-1)^{z\cdot y}\Big(1+(-1)^{a\cdot y}\Big)\Big|^2=\begin{cases}0 & \text{if } a\cdot y=1\\ \frac{1}{2^{n-1}} & \text{if } a\cdot y=0\end{cases}$$

The Hadamard transform creates a uniform superposition over all states $|y\rangle$ such that $y\cdot a=0$. Measurements will always yield uniformly at random some string a such that $y\cdot a=0$.

$$\begin{aligned}y^{(0)}\cdot a&=0\\ &\vdots\\ y^{(n-1)}\cdot a&=0\end{aligned}$$

There are n such equations, one of which is $y=0$. Thus, $n-1$ *linearly independent* equations are needed to solve for a

Given k linearly independent equations, its set of vectors $\{y^{(1)}\dots y^{(k)}\}$ is a basis spanning a subspace of $\mathbb{Z}_2^n$ with dimension 2^k. Consequently, a new equation associated with vector $y^{(k+1)}$, must lie outside of this subspace. That is, it cannot be a linear combination of $\{y^{(1)}\dots y^{(k)}\}$

$$y^{(k+1)}\neq\alpha_1y^{(1)}+\alpha_2y^{(2)}+\cdots+\alpha_ky^{(k)}$$

Its probability of being linearly independent from $\{y^{(1)}\dots y^{(k)}\}$ is:

$$\frac{2^n-2^k}{2^n}=1-2^{k-n}$$

The overall probability of obtaining $n-1$ linearly independent equations is consequently:

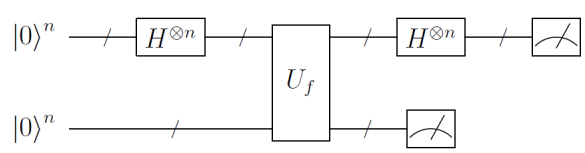
$$\begin{aligned}P&=(1-2^{0-n})(1-2^{1-n})\dots(1-2^{(n-2)-n})\\ &=(1-\frac{1}{2^n})(1-\frac{2}{2^n})\dots(1-\frac{2^{n-2}}{2^n})\\ &=(1-\frac{1}{2^n})(1-\frac{1}{2^{n-1}})\dots(1-\frac{1}{4})\\ &\approxeq 0.2887881\\ &>1/4\end{aligned}$$

We establish the probability's lower bound, by considering $n=\infty$

$$\begin{aligned}P_{min}&=\prod_{n=1}^{\infty}\Big(1-\frac{1}{2^n}\Big)\\ &\approxeq 0.2887881\\ &>1/4\end{aligned}$$

Thus, at most $4n$ runs of the algorithm is necessary to obtain $n-1$ linearly independent equations and solve for a [19] [20] [12].

The quantum circuit for Simon's Algorithm:



7.4 Grover's Algorithm

Problem statement:
The task is to search through an unsorted list of N items $(0, 1, \ldots N-1)$. Mathematically:

$$f(x) = \begin{cases} 0 & \text{if } x \neq x^* \\ 1 & \text{if } x = x^* \end{cases}$$

The task is to find u.

Algorithm:
We assume $N = 2^n$ to store N indexes with n qubits.

Classically, a linear search is required. In the worst-case-scenario, N searches are required if u is the last entry in the list. Grover's algorithm solves this problem in $O(\sqrt{N})$ time [11].

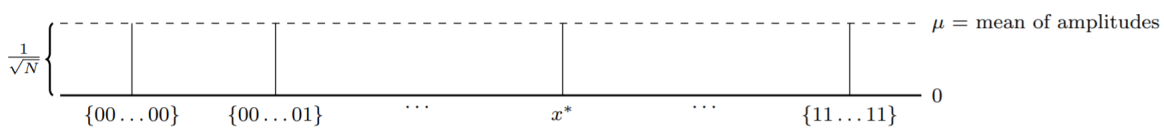
We begin with a query register of n qubits and an answer qubit initialized to $|1\rangle$:

$$|\phi_0\rangle = |0\rangle^{\otimes n} \otimes |1\rangle$$

Applying the Hadamard transform on the query register:

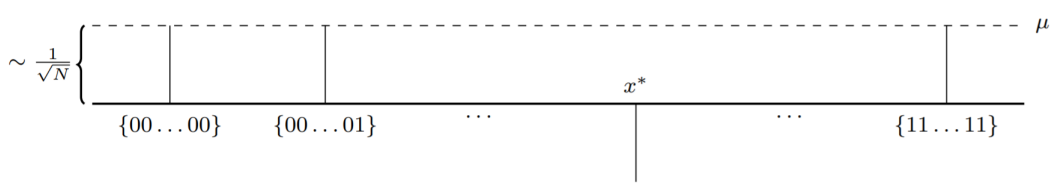
$$\begin{aligned} |\phi_1\rangle &= H^{\otimes n} |0\rangle^{\otimes n} \otimes |1\rangle \\ &= \sum_{x \in \{0,1\}^n} \frac{1}{\sqrt{2^n}} |x\rangle \otimes |1\rangle \\ &= \sum_{x \in \{0,1\}^n} \frac{1}{\sqrt{N}} |x\rangle \otimes |1\rangle \end{aligned}$$

Visually, the amplitudes of the superposition:



For $f(x^*) = 1$, we invert its phase by applying phase kickback, to obtain the superposition

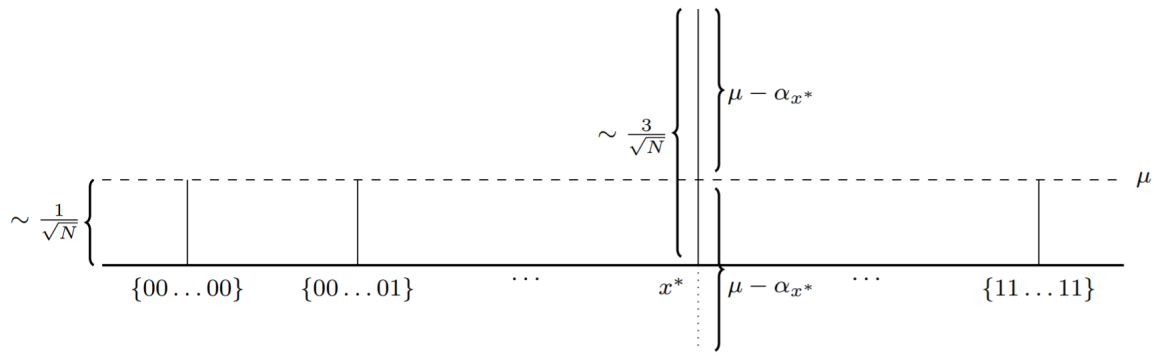
$$|\phi_2\rangle = \sum_{x \neq x*} \alpha_x |x\rangle - \alpha_{x*} |x*\rangle$$



Next, we invert the amplitudes of the superposition about the mean, which we denote μ, to increase the probability of measuring x^*.

$$\mu = \frac{1}{N} \sum_{x=0}^{N-1} \alpha_x$$

Visually, the amplitudes become:



Mathematically, the amplitudes become

$$\begin{aligned}\alpha_x &\to \mu + (\mu - \alpha_x)\\ &= 2\mu - \alpha_x\end{aligned}$$

This inversion is achieved with the *Grover Diffusion Operator*, D, with the following mapping:

$$\sum_{x\in\{0,1\}} \alpha_x \left|x\right\rangle \xrightarrow{D} \sum_{x\in\{0,1\}} (2\mu - \alpha_x) \left|x\right\rangle$$

The Diffusion Operator is mathematically

$$D = H^{\otimes n}\Big(2\left|0\right\rangle\left\langle 0\right| - I\Big)H^{\otimes n}$$

$2\left|0\right\rangle\left\langle 0\right| - I$ is a conditional phase shift that shifts every state except except except $\left|00\ldots0\right\rangle$ by -1

$$\begin{aligned}\Big(2\left|0\right\rangle\left\langle 0\right| - I\Big)\left|0\right\rangle &= 2\left|0\right\rangle\left\langle 0|0\right\rangle - I\left|0\right\rangle\\ &= 2\left|0\right\rangle 1 - \left|0\right\rangle\\ &= 2\left|0\right\rangle - \left|0\right\rangle\\ &= \left|0\right\rangle\end{aligned}$$

$$\begin{aligned}\Big(2\left|0\right\rangle\left\langle 0\right| - I\Big)\left|x\right\rangle &= 2\left|0\right\rangle\left\langle 0|x\right\rangle - I\left|x\right\rangle\\ &= 2\left|0\right\rangle 0 - \left|x\right\rangle\\ &= 0 - \left|x\right\rangle\\ &= -\left|x\right\rangle\end{aligned}$$

[12] Its matrix is:

$$
\begin{aligned}
2\left|0\right\rangle\left\langle 0\right| - I &= 2\begin{bmatrix}1 & 0 & \dots & 0\end{bmatrix}\begin{bmatrix}1\\0\\\vdots\\0\end{bmatrix} - I \\
&= 2\begin{bmatrix}1 & 0 & \dots & 0\\0 & 0 & \dots & 0\\\vdots & \vdots & \ddots & \vdots\\0 & 0 & \dots & 0\end{bmatrix} - I \\
&= \begin{bmatrix}2 & 0 & \dots & 0\\0 & 0 & \dots & 0\\\vdots & \vdots & \ddots & \vdots\\0 & 0 & \dots & 0\end{bmatrix} - \begin{bmatrix}1 & 0 & \dots & 0\\0 & 1 & \dots & 0\\0 & 0 & \ddots & 0\\0 & 0 & \dots & 1\end{bmatrix} \\
&= \begin{bmatrix}1 & 0 & \dots & 0\\0 & -1 & \dots & 0\\\vdots & \vdots & \ddots & \vdots\\0 & 0 & \dots & -1\end{bmatrix}
\end{aligned}
$$

We derive the matrix of D:

$$
\begin{aligned}
D &= H^{\otimes n}\begin{bmatrix}1 & 0 & \dots & 0\\0 & -1 & \dots & 0\\\vdots & \vdots & \ddots & \vdots\\0 & 0 & \dots & -1\end{bmatrix}H^{\otimes n} \\
&= H^{\otimes n}\left(\begin{bmatrix}2 & 0 & \dots & 0\\0 & 0 & \dots & 0\\\vdots & \vdots & \ddots & \vdots\\0 & 0 & \dots & 0\end{bmatrix} - I\right)H^{\otimes n} \\
&= H^{\otimes n}\begin{bmatrix}2 & 0 & \dots & 0\\0 & 0 & \dots & 0\\\vdots & \vdots & \ddots & \vdots\\0 & 0 & \dots & 0\end{bmatrix}H^{\otimes n} - H^{\otimes n} I H^{\otimes n} \\
&= \begin{bmatrix}2/\sqrt{N} & 0 & \dots & 0\\\vdots & \vdots & \ddots & \vdots\\2/\sqrt{N} & 0 & \dots & 0\end{bmatrix}H^{\otimes n} - I \\
&= \begin{bmatrix}2/N & \dots & 2/N\\\vdots & \ddots & \vdots\\2/N & \dots & 2/N\end{bmatrix} - I \\
&= \begin{bmatrix}2/N-1 & 2/N & \dots & 2/N\\2/N & 2/N-1 & \dots & 2/N\\\vdots & \vdots & \ddots & \vdots\\2/N & 2/N & \dots & 2/N-1\end{bmatrix}
\end{aligned}
$$

If we apply D to an arbitrary state $|\alpha\rangle$:

$$D \begin{bmatrix} \alpha_0 \\ \vdots \\ \alpha_x \\ \vdots \\ \alpha_{N-1} \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_x \\ \vdots \\ \beta_{N-1} \end{bmatrix}$$

The amplitudes of $|\alpha\rangle$ are mapped to

$$\begin{aligned} \beta_x &= \frac{2}{N}\alpha_0 + \cdots + (\frac{2}{N} - 1)\alpha_x + \cdots + \frac{2}{N}\alpha_{N-1} \\ &= \frac{2}{N}\alpha_0 + \cdots + \frac{2}{N}\alpha_x - \alpha_x + \cdots + \frac{2}{N}\alpha_{N-1} \\ &= \frac{2}{N}\sum_{x=0}^{N-1}\alpha_x - \alpha_x \\ &= 2\mu - \alpha_x \end{aligned}$$

which is precisely an inversion around the mean.

The Grover iteration needs to be applied $\frac{\pi}{4}\sqrt{N}$ times, which is a quadratic speedup over a classical search algorithm [4].

8 Conclusion

Quantum computing is a quickly emerging technology with the potential of solving problems that even the most powerful supercomputers today cannot begin to scratch the surface of. The algorithms discussed in this essay are an example of this potential.

Nevertheless, it is important to maintain a pragmatic view on quantum computation. Namely, the physical realization of a quantum computer has insofar proven extremely difficult. Quantum decoherence, caused by interactions with the surrounding environment (ex. temperature fluctuations, electromagnetic waves), causes significant computational errors. Today's quantum computers are kept in precisely controlled environments, at temperatures approaching 0 Kelvin. The hardware required to maintain coherence and provide error correction for large-scale computation with many qubits has yet to be developed.

Furthermore, despite the potential impacts of quantum computation in numerous fields like machine learning, and chemistry simulation, the technology also brings substantial ramifications. Quantum computers can break much of today's encryption, which are based on trap-door functions that are computationally infeasible for *classical* computers. As a result, a new era of post-quantum cryptography has emerged, as researchers scramble to find quantum-resistant encryption schemes [21].

The potential of quantum computing is indeed captivating. Tech giants, like Google, and IBM, alongside promising startups such as D-Wave and Rigetti Computing, are all racing to build the first universal quantum computer. One can only wonder, what the state of this technology will be 5, 10, 15 years down the road.

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